

Please ensure this tracking sheet is completed as a group, as it will be used to monitor your individual participation within the team.						
Team Members	Role/ Responsibility	Email				
Emmanuel Oluwajomiloju Akinloye	cleaning and sourcing for data, eigendecomposition, sorting	e.akinloye@alustudent.com				
Garang Buke	PCA and visualization	g.buke@alustudent.com				
Task Allocation and Tracker						
	Task Allocated	Assigned Member	Deadline	Completion Status (Not Started/In Progress/Completed/Did not participate)	Reviewed by Team? (Yes/No)	Comments
	Load African dataset and inspect rows for inconsistent missing values carefully	Emmanuel Oluwajomiloju Akinloye	3/6/2026	Completed	Yes	Dataset loaded successfully and initial inspection clearly identified preprocessing requirements accurately.
	Parse dataset using NumPy functions filtering relevant urbanization indicator values carefully	Emmanuel Oluwajomiloju Akinloye	4/6/2026	Completed	Yes	Relevant urbanization indicators filtered successfully and unnecessary dataset records removed efficiently.
	Clean missing values using column means ensuring complete numerical dataset consistency	Emmanuel Oluwajomiloju Akinloye	5/6/2026	Completed	Yes	Missing values handled successfully and dataset consistency improved for reliable PCA calculations.
	Standardize dataset features using means and standard deviations before PCA calculations	Emmanuel Oluwajomiloju Akinloye	6/6/2026	Completed	Yes	Standardization completed successfully and feature scaling issues eliminated before PCA implementation.
	Compute covariance matrix analyzing relationships and dependencies across dataset variables carefully	Garang Buke	7/6/2026	Completed	Yes	Covariance matrix generated successfully revealing clear feature relationships and redundancy patterns accurately.
	Perform eigendecomposition extracting eigenvalues and eigenvectors from covariance matrix successfully	Garang Buke	8/6/2026	Completed	Yes	Eigenvalues and eigenvectors extracted successfully enabling accurate principal component identification afterward.

	Sort principal components using explained variance percentages for reduction decision making	Garang Buke	9/6/2026	Completed	Yes	Principal components sorted successfully making dimensionality reduction decisions straightforward and effective.
	Project standardized data onto principal components visualizing transformed reduced dimensional distributions	Garang Buke	10/6/2026	Completed	Yes	PCA projection completed successfully and visualization clearly demonstrated dimensionality reduction effectiveness.
	Push notebook to GitHub repository documenting commits and reviewing collaboration comments thoroughly	Garang Buke	11/6/2026	Completed	Yes	SuccessfulFinal notebook uploaded successfully with organized commits and collaboration comments reviewed properly.
	Conduct final assignment review verifying outputs explanations and PCA implementation accuracy	Emmanuel Oluwajomiloju Akinloye	11/6/2026	Completed	Yes	Final review completed successfully ensuring outputs explanations and calculations met assignment requirements.

Meeting Attendance Log

Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps	
3/6/2026	Dataset selection and project planning	Akinloye Emmanuel	Garang , Emmanuel	Discussed suitable African dataset, PCA objectives, and individual task allocation responsibilities.	Begin dataset cleaning and inspect missing values before next meeting.	
5/6/2026	Data cleaning and standardization progress review	Akinloye Emmanuel	Garang , Emmanuel	Reviewed missing value handling, filtering process, and standardization implementation using NumPy functions.	Finalize preprocessing and prepare covariance matrix computation for upcoming session.	
7/6/2026	Covariance matrix and eigendecomposition discussion	Garang Buke	Garang , Emmanuel	Discussed covariance relationships, eigenvalues, eigenvectors, and explained variance interpretation during PCA implementation.	Complete component sorting and begin dimensionality reduction implementation tasks.	

9/6/2026	Principal component selection and visualization review	Garang Buke	Garang , Emmanuel	Evaluated explained variance percentages and compared before and after PCA visualizations results.	Finalize assignment explanations and verify all PCA outputs before submission.	
11/6/2026	Final assignment review and GitHub submission preparation	Akinloye Emmanuel and Garang Buke	Garang , Emmanuel	Reviewed notebook structure, checked outputs, verified explanations, and confirmed successful PCA implementation overall.	Push final notebook to GitHub repository and submit completed assignment files.	